

## Design Rules Verification Report

Filename : C:\Altium\MPPT\_Solar\_Charger006\PCB\_Project\MPPT\_Solar\_Charger.PcbDoc

Warnings 0  
Rule Violations 0  
Waived Violations 2

| Warnings |   |
|----------|---|
| Total    | 0 |

| Rule Violations                                                                                         |   |
|---------------------------------------------------------------------------------------------------------|---|
| Clearance Constraint (Gap=0.254mm) (All),(All)                                                          | 0 |
| Short-Circuit Constraint (Allowed=No) (All),(All)                                                       | 0 |
| Un-Routed Net Constraint ( All )                                                                        | 0 |
| Modified Polygon (Allow modified: No), (Allow shelved: No)                                              | 0 |
| Width Constraint (Min=0.254mm) (Max=1.5mm) (Preferred=1mm) (InNet('3.3'))                               | 0 |
| Width Constraint (Min=0.254mm) (Max=1.5mm) (Preferred=1mm) (InNet('USB_5V'))                            | 0 |
| Width Constraint (Min=0.254mm) (Max=2mm) (Preferred=1mm) (InNet('NetQ5_3'))                             | 0 |
| Width Constraint (Min=0.254mm) (Max=0.254mm) (Preferred=0.254mm) (All)                                  | 0 |
| Width Constraint (Min=0.254mm) (Max=1.5mm) (Preferred=1mm) (InNet('3V3_SYS'))                           | 0 |
| Width Constraint (Min=0.254mm) (Max=2mm) (Preferred=1.5mm) (InNetClass('POWER_HIGH'))                   | 0 |
| Width Constraint (Min=0.254mm) (Max=2mm) (Preferred=1mm) (InNetClass('GND'))                            | 0 |
| Width Constraint (Min=0.254mm) (Max=2mm) (Preferred=1.5mm) (InNet('PV-'))                               | 0 |
| Width Constraint (Min=0.254mm) (Max=0.5mm) (Preferred=0.3mm) (InNetClass('ADC_SENSE'))                  | 0 |
| Width Constraint (Min=0.254mm) (Max=1.5mm) (Preferred=1mm) (InNetClass('POWER_LOW'))                    | 0 |
| Width Constraint (Min=0.254mm) (Max=0.5mm) (Preferred=0.381mm) (InNetClass('USB_DIFF'))                 | 0 |
| Routing Topology Rule(Topology=Shortest) (All)                                                          | 0 |
| Power Plane Connect Rule(Relief Connect)(Expansion=0.508mm) (Conductor Width=0.254mm) (Air Gap=0.254mm) | 0 |
| Hole Size Constraint (Min=0.025mm) (Max=4mm) (All)                                                      | 0 |
| Hole To Hole Clearance (Gap=0.254mm) (All),(All)                                                        | 0 |
| Minimum Solder Mask Sliver (Gap=0.015mm) (All),(All)                                                    | 0 |
| Silk To Solder Mask (Clearance=0.015mm) (IsPad),(All)                                                   | 0 |
| Silk to Silk (Clearance=0.254mm) (All),(All)                                                            | 0 |
| Net Antennae (Tolerance=0mm) (All)                                                                      | 0 |
| Height Constraint (Min=0mm) (Max=25.4mm) (Preferred=12.7mm) (All)                                       | 0 |
| Total                                                                                                   | 0 |

| Waived Violations                                |   |
|--------------------------------------------------|---|
| Hole To Hole Clearance (Gap=0.254mm) (All),(All) | 2 |
| Total                                            | 2 |

| Hole To Hole Clearance (Gap=0.254mm) (All),(All)                                                                    |  |
|---------------------------------------------------------------------------------------------------------------------|--|
| Hole To Hole Clearance Constraint (Collision < 0.254mm) Between Pad CN1-1(119.445mm,27mm) on Multi-Layer And Pad    |  |
| Hole To Hole Clearance Constraint (Collision < 0.254mm) Between Pad CN1-2(111.105mm,27.65mm) on Multi-Layer And Pad |  |